Модуль 2.1

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import tkinter as tk
from tkinter import ttk, messagebox
import psycopg2

BD_CON = {
    "host": "localhost",
    "database": "Demo_3",
    "user": "postgres",
    "password": "okna2301",
    "port": "5432"
}

def get_fio(login):
    con = psycopg2.connect(**BD_CON)
    try:
        cur = con.cursor()
        cur.execute('''
            SELECT name_users, fam_users, otch_users
            FROM users
            WHERE login_users = %s;
        ''', (login,))
        row = cur.fetchone()
        if row:
            return f'{row[1]} {row[0]} {row[2] or ""}'.strip()
    except Exception as e:
        messagebox.showerror("Ошибка", str(e))
    finally:
        cur.close()
        con.close()
    return None

root = tk.Tk()
root.title("Каталог обуви")
root.geometry("1000x750")

f_filtri = tk.Frame(root)
f_filtri.pack(fill="x", padx=10, pady=5)
f_katalog = tk.Frame(root)
f_katalog.pack(fill="both", expand=True, padx=10, pady=5)
c_katalog = tk.Canvas(f_katalog, width=900, height=650, highlightthickness=0)
c_katalog.grid(row=2, column=0, sticky="nsew")
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scroll = ttk.Scrollbar(f_katalog, orient="vertical", command=c_katalog.yview)
scroll.grid(row=2, column=1, sticky="ns")

c_katalog.configure(yscrollcommand=scroll.set)
f_katalog.grid_rowconfigure(2, weight=1)
f_katalog.grid_columnconfigure(0, weight=1)

scroll_frame = tk.Frame(c_katalog)
canvas_frame = c_katalog.create_window((0, 0), window=scroll_frame,
anchor="nw")
scroll_frame.bind("<Configure>", lambda e:
c_katalog.configure(scrollregion=c_katalog.bbox("all")))
c_katalog.bind("<Configure>", lambda e: c_katalog.itemconfig(canvas_frame,
width=e.width))

e_poisk = tk.Entry(f_filtri, width=48)
e_poisk.grid(row=0, column=1, padx=2)
e_poisk.bind("<KeyRelease>", lambda event: katalog())

c_categoria_znachenia = ["Все категории", "Детская обувь", "Менская обувь",
"Мужская обувь"]
c_categoria = ttk.Combobox(f_filtri, values=c_categoria_znachenia, width=13,
state="readonly")
c_categoria.set("Все категории")
c_categoria.grid(row=0, column=2, padx=5)
c_categoria.bind("<<ComboboxSelected>>", lambda event: katalog())
c_zena = ttk.Combobox(f_filtri, values=["мин --> макс", "макс --> мин"],
width=15, state="readonly")
c_zena.set("мин --> макс")
c_zena.grid(row=0, column=3, padx=5)
c_zena.bind("<<ComboboxSelected>>", lambda event: katalog())

def katalog():
    for widget in scroll_frame.winfo_children():
        widget.destroy()

    try:
        con = psycopg2.connect(**BD_CON)
        cur = con.cursor()
        cur.execute('''
            SELECT m.name_models, c.name_categoria, m.img, m.prooizvodstvo,
m.opisanie, m.coctav, m.zena, COALESCE(SUM(mr.count_m), 0) AS count_model
            FROM models m
            LEFT JOIN categoria c ON m.id_categoria = c.id_categoria
            LEFT JOIN model_razmer mr ON m.id_models = mr.id_models
            GROUP BY m.id_models, m.name_models, c.name_categoria, m.img,
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m.prooizvodstvo, m.opisanie, m.coctav, m.zena
            ORDER BY m.id_models ASC;
        ''')
        rows = cur.fetchall()
    except Exception as e:
        messagebox.showerror("Ошибка БД", str(e))
        return

    finally:
        cur.close()
        con.close()
    query = e_poisk.get().strip().lower()

    if query:
        rows = [r for r in rows if query in str(r[0]).lower() or query in
str(r[3]).lower()]
    vybor = c_categoria.get()

    if vybor != "Все категории":
        poisk_slovo = vybor.split()[0].lower()
        rows = [r for r in rows if poisk_slovo in str(r[1]).lower()]
    sort_zena = c_zena.get()

    if sort_zena == "мин --> макс":
        rows.sort(key=lambda r: float(r[6]))

    elif sort_zena == "макс --> мин":
        rows.sort(key=lambda r: float(r[6]), reverse=True)

    for idx, row in enumerate(rows):
        f_pozicia = tk.Frame(scroll_frame, relief="solid", bd=1)
        f_pozicia.grid(row=idx + 3, column=0, pady=(15, 0), sticky="w")
        f_img = tk.Frame(f_pozicia, height=85, width=85, bg="grey")
        f_img.pack_propagate(False)
        f_img.grid(row=0, column=0, padx=5)

        if row[2]:
            try:
                raw_img = tk.PhotoImage(file=f"Images/{row[2]}")
                scale = max(1, raw_img.width() // 85, raw_img.height() // 85)
                img = raw_img.subsample(scale, scale)
                l_img = tk.Label(f_img, image=img, bg="grey")
                l_img.image = img
                l_img.pack(expand=True)

            except Exception:
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                tk.Label(f_img, text="Нет фото", bg="grey",
fg="white").pack(expand=True)

        f_text = tk.Frame(f_pozicia)
        f_text.grid(row=0, column=1, padx=5)
        f_text.grid_columnconfigure(index=0, minsize=650)

        tk.Label(f_text, text=f"{row[3]} | {row[0]}", font=("Arial", 12,
"bold"), anchor="w").grid(row=0, column=0, sticky="w")
        tk.Label(f_text, text=f"Категория: {row[1]}", anchor="w").grid(row=1,
column=0, sticky="w")
        tk.Label(f_text, text=f"Состав: {row[5]}", anchor="w").grid(row=2,
column=0, sticky="w")

        kolvo = row[7]
        kolvo_color = "red" if kolvo < 3 else "black"
        tk.Label(f_text, text=f"На складе: {kolvo} шт.", fg=kolvo_color,
anchor="w").grid(row=3, column=0, sticky="w", pady=2)
        tk.Label(f_text, text=f"Цена: {row[6]} руб.", font=("Arial", 11,
"bold"), anchor="w").grid(row=4, column=0, sticky="w")
katalog()

root.mainloop()
```